

A GUIDE TO PRODUCT QUALITY STANDARDS CARPET & DURRIES



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ABBREVIATIONS

<i>ABCB</i>	Australia Building Codes Board
<i>ANSI</i>	American National Standards Institute
<i>APLAC</i>	Asia Pacific Laboratory Accreditation Co-operation
<i>AS</i>	Australian Standards
<i>ASTM</i>	American Society for Testing and Materials
<i>BIS</i>	Bureau of Indian Standards
<i>CE</i>	Conformité Européenne
<i>CoC</i>	Certificate of Conformity
<i>CPSC</i>	Consumer Products Safety Commission
<i>CPSIA</i>	Consumer Product Safety Improvement Act
<i>CPSA</i>	Consumer Product Safety Act
<i>EC</i>	Council Regulation (Regulation of the European Parliament and Council)
<i>EN</i>	European Standards
<i>FCM</i>	Food Contact Materials
<i>FDA</i>	Food and Drug Administration
<i>FHSA</i>	Federal Hazardous Substances Act
<i>GI</i>	Geographical Indication
<i>GCC</i>	Gulf Co-operation Council
<i>GSO</i>	GCC Standardization Organization
<i>GPSD</i>	General Product Safety Directive
<i>IS</i>	Indian Standard
<i>ISI</i>	Indian Standards Institution
<i>ILAC</i>	International Laboratory Accreditation Co-operation
<i>ISO</i>	International Organization for Standardization
<i>MLA</i>	Multilateral Recognition Arrangement
<i>MRA</i>	Mutual Recognition Arrangement
<i>NABCB</i>	National Accreditation Board for Certification Bodies
<i>NABL</i>	National Accreditation Board for Testing and Calibration Laboratories
<i>SASO</i>	Saudi Standards, Metrology and Quality Organization
<i>QCI</i>	Quality Council of India
<i>QCO</i>	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2,40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This part of the compendium brings relevant regulations and standards pertaining to the carpet and durrie industry. The carpet and durrie industry in Uttar Pradesh, is amongst the key industries and comprises of a major chunk of exports from the state having made their mark in the international markets as products of exceptional quality and workmanship.

WOOLEN & COTTON DURRIE, SITAPUR, JAUNPUR

1. INTRODUCTION

A durrie (also known as dhurri, dhurrie, durry, or durrie) is an Indian term for a hand-woven rug or a thin flat carpet used in home furnishing. The durries have unique designs inspired by their state of origin such as multicolour stripes, which is one of the most popular patterns. Durrie weaving was a big industry in rural India with Durries often used in South Asia as floor-coverings. The concept of durries is a little different from a rug or carpet, because they are used for bedding or packaging and not only as a floor covering. The craft of making woollen carpets or durrie using traditional techniques has been popular since centuries in the region. A range of woollen carpets or durries are made by hand knotting or tufting by the local artisans.



A Durrie is a flat, woven, light rug, usually reversible. It is also lighter as compared to carpet because it is mainly made of cotton, while a carpet uses wool. They have a variety of use depending on size, pattern, and material. The smallest one ranges from a size of 12" by 12" and are used as a table cover for telephone stands, flower vases etc. They are also made in sizes that are ideal for doing meditation 24" by 24", known as an aasan. Durries used in large political or social gatherings may be as large as 20 feet by 20 feet. Durries are easily portable being lightweight and foldable. They come in variety of colour combinations and patterns catering to the needs of any taste or occasion. Durries also have a low maintenance cost.

Jaunpur and Sitapur districts are well known for the production of woollen & cotton durries, in Uttar Pradesh.

2. RAW MATERIALS

Durries are commonly made from four types of materials: cotton, wool, jute, and silk, as well as in variety of combination of all these materials.

To make a woollen carpet, the raw materials used are:

- Wool
- Cotton
- Natural fibres
- Jute fabric

3. MANUFACTURING PROCESS

3.1. Cotton

3.1.1. Yarn Manufacture

The The warp is invariably made of cotton. There are different kinds of cotton threads required for the warp and the weft. The specifications for these threads are, for weft (or baana) six single cotton threads. This remains same for the two qualities of the Durrie produced, regular and stonewashed. For warp (or taana), the quality of the thread varies between the regular product and the stonewashed variety. For regular quality the thread used is 6/6, while for the stonewashed one, it is 12/20.

3.1.2. Rolling

A charkha is used for making rolls or bundles of yarn for the weft.

3.1.3. Thread rolling

The taana machine is made of two basic parts: a big octagonal horizontal cylinder that rotates on its axis, and a vertical frame on which a number of thread rolls can be attached.

3.1.4. Loom frame

Unlike the complex looms used in weaving, the carpet loom has a vertical frame made of two beams (of wood or steel) that are horizontal. The first beam is about 2 feet above the ground and the other is at about 6 feet from the ground. The upper beam is movable and the two beams are tightened by using a screw-and-

chain mechanism. The length of these beams varies depending on the dimensions of the durrie to be woven. The taana or the warp tightened on these beams has two layers that pass through a horizontal metallic frame called the reed. The reed keeps the threads straight and equidistant from each other. Of the two layers of the taana, one remains on the outer side and the other on the inner side. However, this position can be changed using a mechanism called the kamana (a v-shaped wooden frame where the ends are bound with a tight piece of rope) and ruchh (two bamboo pieces on which the kamana is attached with the beam, just above the reed).

3.1.5. Beating

Panja is a metallic, claw-like fork used for beating the weft threads in the warp to adjust it there. The beating is directly proportional to the stability of the durrie.

3.1.6. Finishing

Protruding knots and weft threads are removed using scissors to give the durrie a smooth look. In case of a stonewashed durrie, it is sent to the washer man who washes it using water, detergents, and potassium permanganate. The durrie may develop differential width at the fringes due to shrinkage. If so, it is set tightly on a frame and kept there for a day or two so that it is stretched

properly. It is then sent to the clipper, who clips off all protruding threads and knots using a pair of scissors, giving the durrie a smooth look.

3.1.7. General Quality Checks

Following are a few visual quality parameters/checks:

- The dimension should be as per specifications
- There should be no stains or marks
- There should be no surface wrinkles or folds that cause the carpet to look warped or rippled
- Check for smooth seams at walls and borders

3.2. Wool

3.2.1. Yarn Manufacture

Wool as weft is extensively used in making expensive Durries. There are two types of wool:

- **Hand spun:** This is pure wool procured from the markets of Bikaner and Jodhpur in Rajasthan. This type of wool is used for Durries that are coloured using vegetable dyes. This wool is not of uniform gauge as it is hand spun.
- **Mill-spun:** This too, is pure wool. In Durries made with this kind of wool, normally chemical dyes are used. This is cheaper than hand woven wool and is of uniform gauge.

3.2.2. Thread rolling

The taana machine is made of two basic parts: a big octagonal horizontal cylinder that rotates on its axis, and a vertical frame on which a number of thread rolls can be attached.

3.2.3. Loom frame

Unlike the complex looms used in weaving, the carpet loom has a vertical frame made of two beams (of wood or steel) that are horizontal. The first beam is about 2 feet above the ground and the other is at about 6 feet from the ground. The upper beam is movable and the two beams are tightened by using a screw-and-chain mechanism. The length of these beams varies depending on the dimensions of the durrie to be woven. The taana or the warp tightened on these beams has two layers that pass through a horizontal metallic frame called the reed. The reed keeps the threads straight and equidistant from each other. Of the two layers of the taana, one remains on the outer side and the other on the inner side. However, this position can be changed using a mechanism called the kamana (a v-shaped wooden frame where the ends are bound with a tight piece of rope) and ruchh (two bamboo pieces on which the kamana is attached with the beam, just above the reed).

3.2.4. Beating

Panja is a metallic, claw-like fork used for beating the weft threads in the warp to adjust it there. The beating is directly proportional to the stability of the durrie.

3.2.5. Finishing

Protruding knots and weft threads are removed using scissors to give the durrie a smooth look. In case of a stonewashed durrie, it is sent to the washer man who washes it using water, detergents, and potassium permanganate. The durrie may develop differential width at the fringes due to shrinkage. If so, it is set tightly on a frame and kept there for a day or two so that it is stretched properly. It is then sent to the clipper, who clips off all protruding threads and knots using a pair of scissors, giving the durrie a smooth look.

3.2.6. General Quality Checks

Following are a few visual quality parameters/checks:

- The dimension should be as per specifications
- There should be no stains or marks
- There should be no surface wrinkles or folds that cause the carpet to look warped or rippled

- Check for smooth seams at walls and borders

4. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international (some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (<https://commerce.gov.in/international-trade/trade-agreements/>).

4.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

4.2. International

S. No.	Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) regulation (EC) 1907/2006 b. General Product Safety Directive (GPSD)	a. European Chemicals Agency (ECHA) b. European Commission	a. Hazardous chemicals b. Persistent organic pollutants
2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008 b. Consumer Product Safety Act (CPSA) Act of 1972 c. Federal Hazardous Substances Act (FHSA), Act of 1960.	a. Consumer Product Safety Commission (CPSC)	a. Regulations on chemicals b. Quantity of regulated material c. Product Safety d. Surface flammability
3	Australia	a. Chemicals and Heavy Metals Regulations	a. Australia Building Codes Board (ABCB)	a. Fire Safety regulations for floor covering b. Product safety rules and standards
4	Middle East	a. Saudi Standards, Metrology and Quality Organization (SASO) regulations	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. General product safety regulation b. Labelling requirement c. Hazardous chemicals

5. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects covered	Region/Country	Standard	Aspects covered
General (Refer Table 1)	IS 14466:1997 (Reaffirmed Year 2017)	Fabrics - Description of defects - Vocabulary	-	-	-
	IS 11205: 2011 (Reaffirmed Year: 2020)	Vocabulary of floor coverings	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 11662:1986 (Reaffirmed Year 2019)	Preservative Treatments of Textiles	-	-	-
	IS 673: 1987 (Reaffirmed Year 2022)	Cloth, Woollen, Twill, Dyed	European Union	EN 14041: 2021	Resilient, textile, laminate and modular multilayer floor coverings. Essential characteristics

Domestic Standards			International Standards		
Stage	Code	Aspects covered	Region/ Country	Standard	Aspects covered
	IS 665: 1989 (Reaffirmed Year: 2020)	Textiles - Determination of dimensional changes of fabrics containing wool on soaking in water	USA	ASTM D2462- 13	Moisture in wool
	IS 1349: 1964 (Reaffirmed Year 2017)	Methods for Determination of Wool Fibre Content of Raw Wool	-	-	-
	IS 171:1993 (Reaffirmed Year 2017)	Textiles - Ring spun grey cotton yarn for weaving	-	-	-
	IS 233: Part 1: 2021	Methods for Determination of Length Parameters of Cotton Fibres	-	-	-
	IS 6637: 1972 (Reaffirmed 2018)	Methods for determination of moisture in wool	-	-	-
	IS 3416: 1988 (Reaffirmed Year 2017)	Recommended methods for the removal of non- fibrous matter prior to quantitative analysis of fibre mixtures	-	-	-
	IS 9068: 2021	Recommended methods for the removal of non- fibrous matter prior to quantitative analysis of fibre mixtures	-	-	-
	IS 667: 1981 (Reaffirmed Year 2017)	Methods for Identification of Textile Fibres	-	-	-
Manufacturing/ Processing (Refer Table 3.a & 3.b)	IS 8476: 1977 (Reaffirmed Year: 2020)	Method for determination of wool content in woollen textile materials	European Union	ISO 4919: 2012	Testing: Determination of tuft withdrawal force
	IS/ISO 105:Part X12: 2001	Textiles — Tests for colour fastness — Part X12: Colour fastness to rubbing	European Union	ISO 2550 :1972	Determination of knots
	IS/ISO 105: E01: 2013	Textiles — Tests for colour fastness — Part E01: Colour fastness to water	USA	ASTM D7799- 12a (2021)	Carpet backing adhesives
	IS/ISO 105:B01: 2014	Textiles — Tests for colour fastness — Part B01: Colour fastness to light: Daylight	USA	ASTM D7570-17 (2021) e1	Dimensional stability

Domestic Standards			International Standards		
Stage	Code	Aspects covered	Region/Country	Standard	Aspects covered
	IS/ISO 105: C10: 2006	Textiles — Tests for colour fastness — Part C10: Colour fastness to washing with soap or soap and soda	Australia	AS ISO 2550:1972	Determination of knots
	IS 1315: 1977 (Reaffirmed Year 2018)	Method for determination of linear density of yarns spun on cotton system	Australia	AS ISO 4919: 2012	Testing: Determination of tuft withdrawal force
	IS 681: 2015 (Reaffirmed Year 2019)	Textiles-methods for determination of universal count of woollen and worsted yarn.	Middle East	GSO 537: 1994	Determination Of Number Of Tufts And/or Loops Per Unit Length And Per Unit Area
Finishing (Refer Table 4.a & 4.b)	IS 697: 2005 (Reaffirmed 2020)	Textiles - Druggets (Durries) made from wool, camel hair and goat hair	European Union	ISO 5086: 1977	Sampling and selection of area of test
	IS 681:2015 (Reaffirmed Year: 2016)	Handloom cotton durries (Amalgamating IS 1557)	Australia	AS ISO 5086: 1977	Sampling and selection of area of test
	IS 4905:2015 (Reaffirmed Year: 2020)	Random Sampling and Randomization Procedures (First Revision)	Middle East	GSO ISO 5086: 1977	Sampling and selection of area of test
	IS 17549:2021	Cotton Khadi Durries	-	-	-

5.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 14466:1997 (Reaffirmed Year 2017) - Glossary of terms pertaining to defects in fabrics	Glossary of terms pertaining to defects in fabrics
	IS 11205: 2011 (Reaffirmed Year: 2020) - Textile floor coverings – Vocabulary	Textile floor coverings - Vocabulary

5.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 11662:1986 (Reaffirmed Year 2019) - Preservative treatments of Textiles	Methods to preserve woollen textiles
	IS 673:1987 (Reaffirmed Year 2022) - Textiles - Cloth, Woollen, Twill, Dyed	- This standard prescribes the constructional particulars and other requirements of cloth, woollen, and twill, dyed - This standard does not specify the indeterminable characteristics like general appearance, feel, finish, and shade of cloth

Stage	Code	Scope
	IS 665:1989 (Reaffirmed Year: 2020) - Textiles - Determination of dimensional changes of fabrics containing wool on soaking in water	This standard prescribes a method for determination of dimensional changes of all woven and knitted fabrics containing more than 50 percent wool when wetted out without agitation and dried. The test is applicable to fully finished fabrics
	IS 171:1993 (Reaffirmed Year 2017) - Textiles - Ring spun grey cotton yarn for weaving	Specifies requirements of ring spun (single and doubled) grey cotton yarn for use in weaving
	IS 233: Part 1: 2021 - Methods for determination of length Parameters of Cotton Fibres	Various methods to determine the length of cotton fabrics
	IS 1349:1964 (Reaffirmed Year 2017) - Methods for determination of Wool Fibre Content of Raw Wool	This standard prescribes methods for determination of clean wool fibre content and laboratory scoured yield of raw wool
	IS 6637: 1972 (Reaffirmed 2018) - Methods for determination of moisture in wool	- Test methodology - Referenced documents and adjunct standards
	IS 9068:2021 - Recommended methods for the removal of non-fibrous matter prior to quantitative analysis of fibre mixtures	- Recommended procedures for removal of certain commonly found types of non-fibrous matter - Test methodology and range of results obtained
	IS 667:1981 (Reaffirmed Year 2017) - Methods for Identification of Textile Fibres	Tests for identification of textile fibres
	IS 3416:1988 (Reaffirmed Year 2017) - Method for quantitative chemical analysis of mixtures of polyester fibres with cotton or regenerated cellulose	- Prior testing of materials specification (as per IS 9068:2021, IS 667:1981 (Reaffirmed Year 2017)) - Test Methodology - Adjunct Standards

5.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	ASTM D2462- 13 - Moisture in Wool	Determination of moisture in wool
	GSO ISO 11860: 2015 - Textile floor coverings - Jute carpet backing Fabric	Specifies requirements for primary and secondary jute carpet backing fabrics

5.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	IS 8476: 1977 (Reaffirmed Year: 2020) - Method for determination of wool content in woollen textile materials	Various methods to determine the wool content in carpets
	IS/ISO 105: Part X12: 2001 - Method for determination of colour fastness of textile materials to rubbing	Various test methods to check colour fastness of textile materials to rubbing
	IS/ISO 105: E01: 2013 - Method for determination of colour fastness of textile materials to water	Various test methods to check colour fastness of textile materials to water
	IS/ISO 105: B01: 2014 - Method for determination of colour fastness of textile materials to light	Various test methods to check colour fastness of textile materials to natural light
	IS/ISO 105: C10: 2006 - Method for determination of colour fastness of textile materials to washing	Various test methods to check colour fastness of textile materials to washing

Stage	Code	Scope
	IS 1315: 1977 (Reaffirmed Year 2018) - Method for determination of linear density of yarn spun on cotton system in cotton count and tex	Prescribes methods for determination of linear density of yarn spun on cotton system in cotton count and tex
	IS 681: 2015 (Reaffirmed Year 2019) - Textiles-methods for determination of universal count of woollen and worsted yarn	Prescribes methods for determination of universal count of woollen and worsted yarn

5.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Processing	ISO 4919: 2012 - Testing	Test for Determination of tuft withdrawal force
	ASTM D7799-12a (2021) - Carpet backing adhesive	Standard for carpet adhesives for installation
	ASTM D7570-17 (2021) E1 - Testing	Tests for checking shrinkage of carpet backing fabric
	AS ISO 4919: 2012 - Testing	Test for Determination of tuft withdrawal force
	GSO 537: 1994 -Textile Floor - Coverings - determination of number of Tufts and/or loops per unit length and per unit area	Specifies a method for the determination of the number of tufts and/or loops per unit length and per unit area of a carpet, and is applicable to carpets the pile of which consists of uniformity spaced tufts and/or loops

5.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 697: 2005 (Reaffirmed Year: 2020) - Textiles - Druggets (Durries) made from wool, camel hair and goat hair	Details and other requirements of three varieties of handmade druggets made from wool, camel hair, and goat hair
	IS 681: 2015 (Reaffirmed Year: 2016) - Handloom cotton durries - Specification	Constructional particulars and other requirements for twenty-four varieties of handloom cotton durries
	IS 4905: 2015 (Reaffirmed Year: 2020) - Methods for random sampling	Methods sampling of handmade carpets
	IS 17549: 2021 - Cotton Khadi Durries-specification	Constructional particulars and other requirements for four varieties of handloom cotton khadi durries

5.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	ISO 5086: 1977 - Sampling and Testing	Sampling and selection of areas of test
	AS ISO 5086: 1977 - Sampling and Testing	Sampling and selection of areas of test
	GSO ISO 5086: 1977 - Sampling and Testing	Sampling and selection of areas of test

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement

REFERENCES

1. Product information- <http://odopup.in/en/article/jaunpur>
2. Product information- <http://odopup.in/en/article/sitapur>
3. Domestic Standards and raw materials: <https://standardsbis.bsbedge.com/>
4. ISO Standards: <https://www.iso.org/obp/ui#search>
5. EU: <https://www.en-standard.eu/>
6. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
7. AUS: <https://www.standards.org.au/>
8. MIDDLE EAST: <https://www.gso.org.sa/store/>
9. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

CARPET, BHADOHI, JAUNPUR, SITAPUR, MIRZAPUR AND SONBHADRA

1. INTRODUCTION

A carpet is a textile floor covering typically consisting of an upper layer of pile attached to a backing. The pile was traditionally made from wool, but since the 20th century, synthetic fibres such as polypropylene, nylon or polyester are often used, as these fibres are less expensive than wool. The pile usually consists of twisted tufts that are typically heat-treated to maintain their structure.



The term carpet comes from old French “carpite”. One derivation of the term states that the French term came from the old Italian “carpita”, from the verb “carpire” meaning to pluck.

Carpet weaving may have been introduced to India as far back as the eleventh century with the coming of the first Muslim conquerors, the Ghaznavids and the Ghauris, from the West. It can be traced to the beginning of the Mughal dynasty in the early sixteenth century; the Mughal emperors patronized Persian carpets for their royal courts and palaces. During this period, they brought Persian artisans from

their homeland and established them in India. Initially, these Mughal carpets showed the classic Persian style of fine knotting, and then gradually the style blended with Indian art. Thus, the carpets produced became typical of Indian origin and the industry began to diversify and spread all over the subcontinent. In Uttar Pradesh, Bhadohi, Jaunpur, Sitapur, Mirzapur and Sonbhadra are well known for their prominent carpet manufacturing industry.

The handmade carpets of Bhadohi and Mirzapur are especially renowned the world over and have been given a GI tag in the year 2010 and 2014 respectively, which further enhances their presence in the global markets as a unique product.

2. RAW MATERIALS

The raw materials used for the carpets are essentially:

- Natural fibres
- Wool and wool blends
- Nylon
- Propylene
- Polyester
- Acrylic
- Jute fabric

3. MANUFACTURING PROCESS

Carpets can either be woven or tufted. Based on the production methodology used the fabrics are converted to carpets.

3.1. Tufting Process

There are three main stages in the manufacture of carpets by tufting and a wide variety of fibres can be used:

- a. Tufting:** Tufting begins with the process of weaving the synthetic or staple fibre into a primary backing material. The primary backing material is usually made of woven polypropylene or jute and its main value is to provide a base cloth to hold the yarn in place while the tufting happens. As the needles penetrate the backing, a small hook called a looper grabs the yarn and holds it in place. This process results in what is called loop pile construction.
- b. Application of dye:** The carpet can be either pre-dyed where the individual yarns are dyed before tufting or after tufting. The dyes are prepared traditionally in large vats, in which water is boiled with the chemicals and dyes and the carpet is soaked in to apply the colour. Screen-printing is also done.
- c. Finishing:** This process is typically a single production line that completes the final stage of the carpet construction. In the finishing process, a coating of latex is applied to both the tufted,

dyed carpet's primary backing, and to secondary backing.

Shearing, one of the last stages in the manufacture of carpet is the process of removing all of the little loose ends and projecting fibres that might have been created during the tufting process. Finally, each carpet is carefully inspected for colour uniformity and other manufacturing defects before it is rolled, wrapped, and shipped.

3.2. Weaving Process

The wool is carefully carded and spun into continuous lengths of yarn using a spinning wheel. The yarn is washed, dried, and dyed. The yarn is woven using a loom through or around vertical fibres known as warps, and then locked into place with horizontal strands known as wefts. The highest quality of carpet is achieved with the woven carpet manufacturing. This process creates a variety of rich colours and intricate patterns. Carpets are primarily woven of wool and are very durable. The knots are made manually and are a laborious process. The higher the number of the knots, the finer the quality of the rug. The readied rug is washed and then using shears, finishing is done.

4. GENERAL QUALITY CHECKS

Following are a few visual quality parameters/checks:

- Check the pile height to the nearest millimetre
- The dimension should be as per specifications
- There should be no stains or marks
- There should be no surface wrinkles or folds that cause the carpet to look warped or rippled
- Check for smooth seams at walls and borders

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations, domestic and international

(some key regions/countries), are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (<https://www.fieo.org/>)
- Directorate General of Foreign Trade (<https://www.dgft.gov.in/CP/>)
- Export Promotion Councils (<https://commerce.gov.in/useful-links/export-promotion-councils/>)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce

(<https://commerce.gov.in/international-trade/trade-agreements/>).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

5.2. International

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
1	European Union	a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulation (EC) 1907/2006 b. General Product Safety Directive (GPSD)	a. European Chemical Agency (ECHA) b. European Commission.	a. Hazardous chemicals b. Persistent organic pollutants

S. No.	Country	Regulation Reference	Regulatory Body	Regulation
2	United States of America	a. Consumer Product Safety Improvement Act (CPSIA), Act of 2008 b. Consumer Product Safety Act (CPSA), Act of 1972 c. Federal Hazardous Substances Act (FHSA), Act of 1960.	a. Consumer Product Safety Commission (CPSC)	a. Surface flammability (3-Oct, 2000) b. Consumer Product Safety Improvement, Act 2008
3	Australia	a. Fire Safety Regulations	a. Australia Building Codes Board (ABCB)	a. Fire Safety regulations for floor covering, Building code Australia b. Product safety rules and standards
4	Middle East	a. Saudi Standards, Metrology and Quality Organization (SASO) regulations	a. Saudi Standards, Metrology and Quality Organization (SASO)	a. Labelling requirements b. Hazardous chemicals

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
General (Refer Table 1)	IS 11205: 2011 (Reaffirmed Year: 2020)	Vocabulary of textile floor coverings	-	-	-
Raw Materials (Refer Table 2.a & 2.b)	IS 4900: PART 1 TO 3: 1984 (Reaffirmed Year: 2021)	Jute carpet backing fabric	European Union	EN 1307: 2014	Classification of textile floor coverings
	IS 5910:1977: 1984 (Reaffirmed Year: 2022)	Fineness grades of wool	USA	ASTM D7799 -12a (2021)	Carpet backing adhesive
	IS 744:2000 (Reaffirmed Year: 2022)	Textiles - Methods for Determination of Wool Fibre Diameter, Percentage of Medullated Fibres and Kemp Fibre (Amalgamating IS 1348 and 2899)	Australia	ISO 11860 :1999	Jute backing guidelines
	IS 9393:1979 (Reaffirmed Year: 2022)	Manufacturing worsted, semi-worsted and woollen yarns and tolerances	-	-	-

Domestic Standards			International Standards		
Stage	Code	Aspects Covered	Region/ Country	Standard	Aspects Covered
	IS 10921:1984 (Reaffirmed Year: 2020)	Specification for carpet yarn made from virgin wool	-	-	-
Manufacturing/Processing (Refer Table 3.a & 3.b)	IS 11662:1986 (Reaffirmed Year 2019)	Preservative treatment of textiles	Australia	ISO 11859:1999	Textile floor coverings, hand knotted woollen carpet
	IS 5884:2020	Constructional and performance requirements for Hand tufted Carpets	Middle East	GSO 857:2007	Standards for woven carpets
	IS 5641:1993 (Reaffirmed Year: 2020)	Textile floor covering - Hand-made wool carpets	European Union	EN 14041:2021	Essential characteristics for the following types of floor coverings
	-	-	USA	ASTM D7983-17	Shrinkage of carpet backing fabric
	-	-	USA	ASTM D7570-17 (2021) e1	Dimensional stability of pile yarn floor covering
	-	-	-	-	-
Finishing (Refer Table 4.a & 4.b)	IS 7877: Part1 TO 5: 1976 (Reaffirmed Year: 2022)	Methods of sampling and tests for Hand-made carpets	Australia	AS/NZS 3733 :2018	Maintenance of carpets
	IS 8476:1977 (Reaffirmed Year: 2020)	Determination of wool content in carpets	Australia	ISO 4919 :2012	Determination of tuft withdrawal force
	-	-	Australia	ISO 2550: 1972	Determination of knots
	-	-	Australia	ISO 5086:1977	Sampling and selection of areas of test
	-	-	Middle East	GSO 537:1994	Determination Of Number Of Tufts and/or loops per unit length and per unit area
	-	-	USA	ASTM D7983-17	Shrinkage of carpet backing fabric

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	IS 11205: 2011 (Reaffirmed Year: 2020) - Vocabulary of textile floor coverings	Definitions of terms relating to textile floor coverings, and defining categories of these products

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	IS 5910:1977: 1984 (Reaffirmed Year: 2022) - Fineness grades of wool	- Classification of fineness grades of the fibres in raw wool. - Test methodology - Assignment of grades - Adjunct Standards
	IS 744:2000 (Reaffirmed Year: 2022) - Textiles - Methods for determination of Wool fibre diameter, Percentage of Medullated fibres and Kemp fibre (Amalgamating IS 1348 and 2899)	- Terminology - Methods of tests - Adjunct standards and referenced documents
	IS 10921:1984 (Reaffirmed Year: 2020) - Specification for carpet yarn made from virgin wool	- Manufacturing details and other requirements of wool yarn - Terminology and tests
	IS 9393:1979 (Reaffirmed Year: 2022) - Manufacturing worsted, semi-worsted and woollen yarns and tolerances	- Sequence of processes involved in the manufacturing of worsted, semi-worsted, and woollen yarn. - Methods of tests
	IS 4900: PART 1 TO 3: 1984 (Reaffirmed Year: 2021) - Jute carpet backing fabric	General requirements and information regarding criteria for conformity of the jute carpet backing fabric woven in both plain and sateen weaves

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	EN 1307: 2014 - Requirements for classification, domestic or commercial, of all carpets and carpet tiles, excluding rugs and runners into use classes	Specifies the requirements for classification, domestic or commercial, of all carpets and carpet tiles, excluding rugs and runners into use classes
	ASTM D7799-12a (2021) - Standard Specification for tufted and woven broadloom carpet Adhesives without homogenous PVC or Non-PVC backings	- Establishes minimum requirements for adhesives used for the installation of tufted or woven broadloom carpet, within a climate-controlled structure, when adhered directly, and permanently, to a structurally sound and recommended substrate. - Provides the means to determine adhesive bonding variation because of different carpet backing systems and their usage classifications based on the traffic conditions they are subjected to
	ISO 11860:1999 - Requirements for primary and secondary jute carpet backing fabrics	Specifies requirements for primary and secondary jute carpet backing fabrics

6.4. Table 3.a: (Domestic Standards for Processing)

Stage	Code	Scope
Manufacturing	IS 11662:1986 (Reaffirmed Year 2019) - Preservative treatments of textiles	Specifies treatments of common preservatives used in the textile industry for protection against one or more causes, namely attack by fungi, bacteria, and insects and degradation caused by sunlight and weathering

Stage	Code	Scope
	IS 5884:2020 - Textile floor covering — hand tufted carpets — specification	Standard prescribes the constructional particulars and performance requirements of tufted carpets made from wool or man-made fibres or their blends.

6.5. Table 3.b: (International Standards for Processing)

Stage	Code	Scope
Manufacturing	ISO 11859:1999 - Textile floor coverings — pure wool, hand-knotted pile carpets — specification	Standard specifies requirements for hand-knotted carpets produced from pure wool, of dimensions agreed between the purchaser and the supplier
	GSO 857: 2007 - Woven Carpets	Standard specifies woven carpets manufactured by the use of single or face-to-face traditional type looms, or other type of carpet weaving machinery
	EN 14041:2021 - Resilient, textile, laminate and modular multilayer floor coverings. essential characteristics	Specifies the essential characteristics for different types of floor coverings
	ASTM D7570-17 (2021) e1 - dimensional stability of pile yarn floor covering	Tests for checking shrinkage of carpet backing fabric
	ASTM D7983-17 - Shrinkage of carpet backing fabric	Test methods cover the procedure for evaluating shrinkage, in both wet and dry conditions, of woven, knitted and nonwoven backing fabrics designed for use in the manufacture of pile yarn floor coverings

6.6. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	IS 7877: Part1 TO 5: 1976 (Reaffirmed Year: 2022) - Methods of Sampling and tests for hand-made Carpets	Standard (Part I) gives the method for sampling and selection of areas for carrying out the physical tests and chemical analysis of hand-made carpets
	IS 8476:1977 (Reaffirmed Year: 2020) - Method for determination of wool content in woollen textile materials	Various methods to determine the wool content in carpets

6.7. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	AS/NZS 3733 :2018 - Maintenance of carpets	Provides requirements, techniques, and guidelines for cleaning maintenance of residential and commercial textile floor coverings
	ISO 4919 :2012 - Determination of tuft withdrawal force	Method for determination of tuft withdrawal force and is applicable to all carpets with a cut or loop pile yarn structure
	ISO 2550: 1972 - Determination of knots	Describes different types of knots used in manufacturing hand-made carpets. A visual method for identification of types of knots is given. Applicable to most carpets in which knots are tied by finger or by hook
	ISO 5086 – 1977 - Sampling and selection of areas of test	Specifies the methods for sampling and defines the areas for the physical testing and chemical analysis of hand-knotted carpets

Stage	Code	Scope
	GSO 537 – 1994 - Determination of Number of Tufts and/or Loops per Unit Length and Per Unit Area	Specifies a method for the determination of the number of tufts and/or loops per unit length and per unit area of a carpet
	ASTM D7983-17 - Shrinkage of carpet backing fabric	Test method covers the procedure for evaluating shrinkage, in both wet and dry conditions, of woven, knitted, and nonwoven backing fabrics designed for use in the manufacture of pile yarn floor coverings

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

1. Product information: <http://odopup.in/en/article/bhadohi>
2. Product information: <http://odopup.in/en/article/jaunpur>
3. Product information: <http://odopup.in/en/article/sitapur>
4. Product information: <http://odopup.in/en/article/mirzapur>
5. Product information: <https://strathmoredesign.com/carpethowitsmade>
6. Product information: <https://en.wikipedia.org/wiki/Carpet>
7. International Legislation/Regulation:
 - a. EU: https://circulareconomy.europa.eu/platform/sites/default/files/knowledge_-_toxics_in_carpets_eu_review_anthesis_final_study.pdf
 - b. USA: <https://www.federalregister.gov/documents/2000/03/>
 - c. AUS: https://www.carpetinstitute.com.au/wp-content/uploads/2014/02/Carpets091v3_Fire_FactSheet.pdf
8. Domestic Standards for Carpets and raw materials: <https://standardsbis.bsbedge.com/>
9. ISO Standards: <https://www.iso.org/obp/ui#search>
10. International Standards:
 - a. EU: <https://www.en-standard.eu/>
 - b. USA: <https://www.en-standard.eu/astm-standards/>, <https://www.astm.org/>
 - c. AUS: <https://www.standards.org.au/>
 - d. MIDDLE EAST: <https://www.qso.org.sa/store/>
11. NABL accredited laboratories: <https://nabl-india.org/>, <https://parakh.ncog.gov.in>

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials,

moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products,

services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such

products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL):** <https://nabl-india.org/>
- **Bureau of Indian Standards (BIS):** <https://bis.gov.in/>

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an

organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the

relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National

Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB):
<http://nabcb.qci.org.in/>

APPENDIX II: IMPORTANT INSTITUTIONS

CARPET & DURRIES				
S. No.	Institute	Areas of Expertise	Postal Address	Contact
1	Indian Institute of Carpet Technology	- Carpet Technology - Textile Design Technology - Home Textile Technology	Indian Institute of Carpet Technology , Bhadohi, Chauri Rd, Bhadohi Nagar Palika, Uttar Pradesh 221401	Email: pstodirector@iict.c.in Phone: 0542-2307047 Website: www.iict.ac.in
2	Indian Institute of Technology- Banaras Hindu University (IIT-BHU) School of Material Sciences and Technology	- Polymer Nano-composites - Biodegradable Polymers	IIT(BHU), Ajagara, Varanasi, Uttar Pradesh 221005	Email: coordinator.mst@iitbhu.ac.in Phone: 0542-2307047 Website: www.iitbhu.ac.in/dept/mst

